

Progress Report Sprint 9

- Course Name: CS461 Senior Software Engineering Project I
- Date: 02/25/26
- Team Name: TaxonBodyMassML
- Team ID: CS.032
- Student Names and Emails:
 - Grant Pasquantonio (pasquang@oregonstate.edu)
 - Hailey Prater (praterh@oregonstate.edu)
 - Anish Chowdary Pendurti (pendurta@oregonstate.edu)

Summary - Outcomes, blockers, and insights

- **UI: Made informed changes to the UI based off of interviews; plugged content into the 'Help' page and updated the "More Questions" functionality**
- **Completed taxonomy data second pass using the NCBI API, successfully compiling about 37,000 species taxonomies.**
- **Retrained the decision tree regressor on updated taxonomy dataset.**
- **Met last sprint's goal to complete more work on the 'Help' page while still putting in work on the algorithm front**

Blockers:

- **Realized that the UI wasn't as friendly toward people of lower technical literacy as we thought; it needed a couple updates to accommodate this realization.**

Work Log

1. Interviewing and Subsequent UI and 'FAQ' Content Updates

Status: Complete

Owner: Hailey

Description: In order to meet 'Help' page goals from last sprint, I interviewed three people of varying technical literacy levels to 1) find places to improve usability and 2) plug some content in the 'FAQ' section. I took notes during the interviews and used those notes to make my updates. Watching a third party interact with the UI also allowed me to discover some small UI bugs that I have since fixed.

Source Link: [w 2026-02-12-CS.032-Sprint8-ProgressReport.docx](#)

Supporting Artifact: Item.01 and Item.02

Notes: I definitely want to do some more interviewing in the future, but this round of interviewing and updating is considered "complete".

2. Help Page 'More Questions' Functionality

Status: In-progress

Owner: Hailey

Description: I have updated the 'More Questions' chatbox on the 'Help' page to show questions permanently between systems. I did this by introducing another API call through render called "questions" instead of "single_species". A json file is created, updated, and frequently re-accessed to populate the questions.

Source Link: [W 2026-02-12-CS.032-Sprint8-ProgressReport.docx](#)

Supporting Artifact: Item.03

3. Decision Tree Regression

Status: Completed

Owner: Grant

Description: The final version of the Decision Tree Regression has been completed. The model size seems to adequately match the size of the dataset. Future steps will involve integrating the model into a microservice such that it can make predictions upon requests from the frontend.

Source Link: [2026-01-13-CS.032-Sprint6-ProgressReport.docx](#)

Supporting Artifacts: Item.04

Notes: For information about the XGBoost library, check out the manpages here: [XGBoost](#)

4. Taxonomy Data 2nd Pass

Status: Completed

Owner: Grant

Description: During the first attempt at gathering taxonomy data using an API, there were many missing values, particularly from the class and order categories. As such, we came up with the solution of using another taxonomy data API to cross reference to fill in any missing values. This was completed using the NCBI API.

Source Link: [2026-01-30-CS.032-Sprint7-ProgressReport.docx](#)

Supporting Artifacts: Item.04

Notes: For information about the API, check out the manpages here: [NCBI API](#)

5. API Testing and Calling

Status: In-progress

Owner: Anish

Description: The code for the API is almost complete, we just to pass some instructions into the API and see if the json objects are being generated from the data set

Source Link: [2026-01-30-CS.032-Sprint7-ProgressReport.docx](#)

Supporting Artifacts: Item.04

Notes: For information about the API, check out the manpages here: [NCBI API](#)

Risk and Quality Tracking

Open Bug Count: 3

TOP BUGS	SEVERITY	OWNER(S)	STATUS	LINK	ACTIONS
DATASET TOO LARGE FOR DECISION TREE	High	Grant	In-progress	Decision Tree Code on GitHub	Since the model worked with about 30% of the dataset, as soon as the rest of the data is fixed, Grant will try running the model on a larger portion of the data.
NON-RESPONSIVE CSS	Low	Hailey	In-progress	primary CSS file on GitHub	Hailey will add in CSS that better accommodates smaller screen sizes (media queries) by the end of this term (not a top priority)
MISSING CLASS/ORDER FROM MANY DATASET ROWS	High	Grant	Completed	Data Combination Code on GitHub	Successfully used NCBI API to fill in many of the missing rows of data in the taxonomy dataset. About 37,000 of 42,000 rows are now filled. For now, we will focus on other higher priority issues.

Next Sprint Plan

- We need to incorporate the new algorithm into the front-end. This means altering the API requests to handle the new microservice instead of the look-up table microservice. This works toward pretty much every single functionality-based requirement from the beginning of last term.
Priority: High; **Link:** [2025-10-23-CS.032-Requirements](#)
- We need to complete a functional version of the LLM powered predictive model. For now, this will involve using strictly taxonomy data. In future versions, we will incorporate phylogenetic data using This is a prerequisite for integrating the model into the front-end system. As such, it is necessary to fulfill REQ-008 involving accuracy on predictions.
Priority: High; **Link:** [2025-10-23-CS.032-Requirements](#) (REQ-008)

Team Reflection

This sprint was very productive on the predictive algorithm front. We now have a regression-based implementation that we consider “complete”. After converting it to a microservice, we can integrate it into the front-end very easily; this is our primary goal for the last sprint of the term, and one that we need to get done fairly quickly in order to present it during lecture.

As for last sprint’s primary goal, the ‘Help’ page saw vast improvement; not only was the ‘FAQ’ section populated with relevant information, but the ‘More Questions’ feature has also been edited so that questions populate between systems. Through the interviewing process that helped with ‘FAQ’ content, the usability and quality behind our application also benefitted.

Overall, we are excited to have an algorithm to implement in the front-end because we have been striving to do so since the beginning of the project. This has reinvigorated our desire to do our best work in the last sprint of this term and throughout all of the next.

Artifacts Appendix

Item.01

TaxonBodyMass Third-Party UI Interview

This is the document used to track questions and recommendations from the three people interviewed; everything on the document was addressed in some way during this sprint.

Item.02

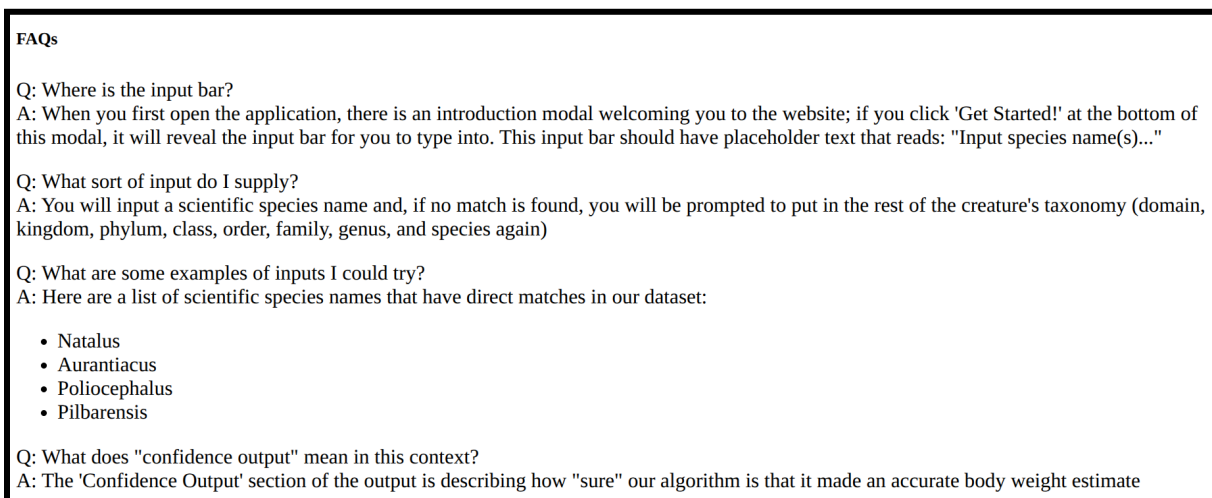


Fig.01 - This is the updated ‘FAQs’ section of the ‘Help’ page after plugging in the questions from the conducted interviews. These align with the interview documentation (Item.01).

Item.03

[Video of Question Permanence Between Systems](#)

This video depicts the updated 'More Questions' feature of the 'Help' page. The questions are no longer stored locally, and are visible between systems. This is shown by the question also appearing on the web page when it is brought up by the private browser.

Item.04

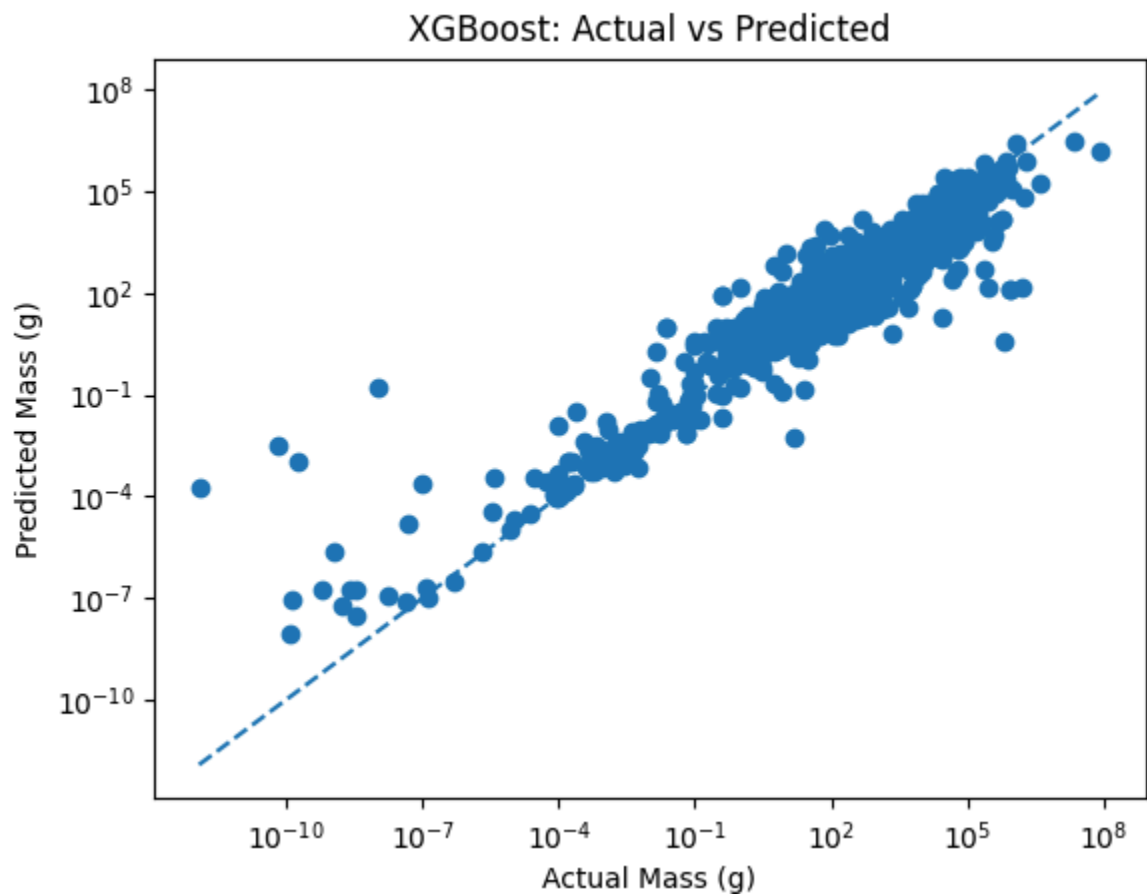


Fig.03 - This is a log-scale plot of the mass predictions for species in the test dataset. This has been updated since the last sprint because the dataset size has been nearly tripled due to the taxonomy dataset second pass.